A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Manipulation of food species & agricultural energetics - Set 1

Mark scheme

AQA spec 3.5.1.2; Biological resources; textbook pp. 321-327

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Selective breeding:** parents chosen for desirable genetic characteristics (Limousin muscle growth, Merino wool, Cheviot hardiness) – combines genes but characteristics of offspring not fully predictable – many generations needed – risk of inbreeding **2** Asexual reproduction (cuttings, runners, tissue culture, cloning): offspring genetically identical to the parent so characteristics are predictable and survival high – but no variation so no improvement and vulnerability to the same disease – cloning of animals (nucleus transfer into an enucleated egg, surrogate) still developing **2 Cross-breeding:** two breeds combined to give a mix of characteristics with hybrid vigour (heterosis) and lower inbreeding risk, eg Zebu (heat tolerant) x Ayrshire (high milk yield) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	6

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 4 points below. **1 AI:** semen from a male with desirable characteristics can fertilise many more females than natural mating – frozen semen stored and transported easily and used after the male dies **1 Embryo transfer:** FSH stimulates a superior female to release many eggs, which are fertilised (IVF) with sperm from a superior male **1** Each embryo is implanted in a different surrogate female **1** So the donor produces many offspring per year rather than one per pregnancy **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-323; AQA spec 3.5.1.2.	
02.1	**1 mark each** 1 mark for each of the 2 points below. **1** Single genes can be transferred from any species, not just closely related ones that produce fertile hybrids **1** Desired characteristics introduced without linked unwanted ones – faster **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	2
02.2	**1 mark each** 1 mark for each of the 3 points below. **1 Bt:** reduces insecticide spraying so fewer non-target deaths, lower cost / energy **1 Roundup Ready:** weeds controlled by spraying glyphosate without harming the crop, so higher yields with less cultivation / tillage **1 Golden rice:** provides vitamin A in the staple diet, reducing about 500 000 cases of blindness a year in India / sub-Saharan Africa **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

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Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 4 points below. **1** Gene transfer by pollen to conventional / organic crops or wild relatives (eg Bt gene killing insects feeding on wild plants) **1** Antibiotic-resistance marker genes could transfer to pathogenic bacteria – possible allergies – gene transfer to gut bacteria **1** Detection in the human food chain from crops approved only for animal feed **1 Economic:** seeds must be bought each year from patent-holding companies, raising costs and reducing local crop diversity and seed saving in LEDCs – herbicide-resistant crops may increase herbicide use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use	4
02.4	1 mark for the point below. **1** Public / political concern that risks are not fully assessed - more evidence needed (precautionary principle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 323-324; AQA spec 3.5.1.2.	
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Extensive cereal 5.0 – intensive cereal 3.0 **1** Intensive beef 0.1 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. **Do not credit** • A bare wrong answer with no working.	2

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Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0) **1** Each extra unit of input gives a smaller extra yield (law of diminishing returns) **1** Beef has very low productivity and efficiency because energy is lost at an extra trophic level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
03.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Any input that increases productivity but requires the use of energy (usually fossil fuel) **up to 2** Manufacture of nitrate fertiliser (Haber process) and pesticides – fuel for machinery – pumping irrigation water – grain drying – food processing and transport – embodied energy of machinery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases **1** Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.	

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Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) **1** Fish are ectotherms so no energy is used for thermoregulation and water supports their mass – poultry grow fast and are kept warm indoors with movement restricted **1** Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Control of movement (housing / restricted grazing) so less energy is used in muscle activity **1** Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature **1 Species selection:** choose species with low FCRs (poultry, fish) – high-energy easily digested diets – slaughter at the fastest growth stage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Only about 10% of the energy at one trophic level is passed to the next – most is lost as heat in respiration / metabolism or in undigested material **1** Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) **1** Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot **1** Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	

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Question	Marking guidance	Marks
05.1	1 mark for the point below. **1** Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Zebu tolerate heat but have a low milk yield – Ayrshire have a high milk yield but are not heat tolerant **1** Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2
05.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Replacement of valuable animals that die with genetically identical individuals **1** Replacement of herds culled during a disease outbreak **1** Production of large numbers of individuals with desirable characteristics **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
06.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1000 / 100 000 x 100 = 1% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

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Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some light is reflected by the leaves **1** Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed **1** Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) **1** After four levels too little energy remains to support a further population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) **1** Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
06.5	**1 mark each** 1 mark for each of the 2 points below. **1** Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) **1** So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	

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Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Yield increases as inputs increase **1** But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Second unit:** 70 - 50 = 20 units **1 Fifth unit:** 94 - 90 = 4 units **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (units) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** The first units of input on each area give the largest yield increases **1** Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Some areas have much greater potential productivity so investment there is sensible **1** Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.5	**1 mark each** 1 mark for each of the 2 points below. **1** Yield per unit area is very high because of large inputs **1** But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	

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